Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
CS101 - PROGRAMMING FOR PROBLEM SOLVING

TIME ALLOTTED: 3Hours **Maximum Marks : 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. float occupies how many bytes in memory **(1)** CO1 BL1 PO1

   (a) 4 (b) 8
   (c) 10 (d) 12

2. What is the output of the following code – **(1)** CO1 BL1 PO1

```
#include <stdio.h>
int main() {
int i = 2;
switch (i) {
case 0:
printf("Hello");
break;
case 1:
printf("World");
break;
default:
printf("HelloWorld");
```

```
        }
        return 0;
}
```

1. Hello 2. World 3. HelloWorld 4. Error

(a) Hello (b) World
(c) HelloWorld (d) Error

3. What will be the output of this code? **(1)** CO1 BL1 PO1

```
int myarray[] = {5, 5, 75, 10};
printf("%d", myarray[0]);
```

(a) 5 (b) 6
(c) 4 (d) none of the above

4. What is the output of the following code – **(1)** CO1 BL1 PO1

```
# include <stdio.h>
void myfunction(int x)
{
        x = 20;
}
int main()
{
        int y = 10;
        myfunction(y);
        printf("%d", y);
        return 0;
}
```

(a) 10 (b) 20
(c) Compile Error (d) None of these

5. What header file is used for string operation? **(1)** CO1 BL1 PO1

(a) <string.h> (b) <stdio.h>
(c) <stdlib.h> (d) <math.h>

6. Which of the following is a ternary operator? **(1)** CO1 BL1 PO1

(a) ?: (b) *

(c) sizeof (d) ^

7. Which of the following are tokens in C? **(1)** CO1 BL1 PO1

(a) a) Keywords (b) b) Variables

(c) c) Constants (d) d) All of the above

8. Which function dynamically allocates memory? **(1)** CO3 BL2 PO2

(a) free() (b) fopen()

(c) malloc() (d) exit()

9. Which loop executes at least once? **(1)** CO2 BL3 PO4

(a) while (b) for

(c) do-while (d) None

10. What is printed by this code? **(1)** CO3 BL3 PO6

```
int x = 20;
int *p = &x;
printf("%d", *p);
```

(a) Address of x (b) 20

(c) 0 (d) Error

11. How many times will the loop execute? **(1)** CO2 BL3 PO2

```
for(int i = 1; i <= 5; i++)
```

(a) 3 (b) 4

(c) 5 (d) 6

12. To create a program for storing student details, which feature is MOST suitable? **(1)** CO5 BL6 PO3

(a) Array (b) Structure

(c) enum (d) Pointer

Part B

(Answer any three of the following) 3x5=15

13. Write a C program to find the length of a string without using string.h header file. **(5)** CO2 BL2 PO2

14. What are the advantage of 2's complement number over 1's complement number? **(5)** CO3 BL3 PO5

Convert (123.7)10 into equivalent binary number.
(2+3)

15. Discuss different basic data types in C. Provide an example for each.	**(5)**	CO2	BL2	PO1
16. Explain static vs dynamic memory allocation in C. Mention one use-case of malloc().	**(5)**	CO4	BL3	PO5
17. Write a C program using switch-case to implement a menu-driven calculator (min. 5 operations).	**(5)**	CO3	BL4	PO2

Part C
(Answer any three of the following) 3x15=45

18. i. Differentiate between while and do-while statements with suitable examples. ii. Explain with an example what is a function in C. What is the utility of using function in C. iii. Write a C program to generate Fibonacci series. (5+5+5)	**(15)**	CO3	BL3	PO6
19. Write a short note on any three i. C pre-processor ii. Structure and Union iii. Switch-Case in C iv. Pointer Arithmetic v. Bitwise Operator (5+5+5)	**(15)**	CO2	BL6	PO4
20. a) Define recursion. Differentiate between iteration and recursion. (2+2) b) Explain call by value & call by reference with suitable programs. (3+3) c) Write a C program to delete a specific line from a file. (5)	**(15)**	CO3	BL4	PO3
21. Answer all the questions				
(a) Explain memory representation of 2D arrays in C (row-major order) with an example.	**(5)**	CO4	BL3	PO5

	(b) Write a C program to perform matrix multiplication of two compatible matrices.	**(5)**	CO4	BL4	PO5
	(c) Evaluate two advantages and two limitations of using arrays for data storage.	**(5)**	CO4	BL5	PO5
22.	(a) Explain text vs binary files and formatted vs unformatted I/O.	**(5)**	CO4	BL3	PO5
	(b) Write a program to create a text file and store a sentence entered by user; then read and display it.	**(5)**	CO4	BL5	PO5
	(c) Extend the program to append another sentence and display final file content.	**(5)**	CO4	BL6	PO5